Live View Overview

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Live View is the core real-time monitoring feature in GCXONE, providing instant access to live video feeds from all connected cameras and devices. It enables operators to monitor sites in real-time, respond to events immediately, and maintain situational awareness across multiple locations.

Key Capabilities

Real-Time Video Streaming

- Instant access to live video feeds with minimal delay
- Support for high-resolution video streams
- Automatic quality adjustment based on bandwidth
- Support for RTSP, P2P, and manufacturer-specific protocols

Multi-Camera Monitoring

- View multiple cameras simultaneously in organized grid layouts
- Create and save custom camera arrangements
- Switch between single camera and multi-camera views
- Support for panoramic camera views

Interactive Controls

- PTZ Control: Direct pan-tilt-zoom control from live view on supported cameras
- Audio Controls: Two-way audio communication on supported devices
- Preset Access: Quick access to camera preset positions
- Full-Screen Mode: Expand camera to full screen for detailed viewing
- Snapshot: Capture still images from live video

Performance Features

- Local Mode: Direct P2P streaming for reduced latency
- Intelligent bandwidth and quality management
- Support for multiple simultaneous streams
- Automatic stream refresh and reconnection

How It Works

When a user selects a camera for live view, GCXONE establishes a connection to the camera or NVR, initiates the video stream, and delivers it to the user interface. The stream is maintained continuously for real-time monitoring.

Deployment Models

Cloud Streaming: Video routed through GCXONE cloud infrastructure. Accessible from anywhere with internet. Encrypted, secure, and optimized for remote access.

Local Mode (P2P): Direct connection between workstation and device. Reduced latency for on-site monitoring, lower bandwidth on WAN, and enhanced performance for operator workstations.

Supported Devices

- IP Cameras: All supported camera manufacturers
- NVRs: Network video recorders from all supported vendors
- VMS Platforms: Milestone, Avigilon, Digital Watchdog, Genesis VMS, and more
- Cloud Devices: NXG Cloud NVR, NXG Cloud Vision Edge
- Specialized surveillance devices

Use Cases

Real-Time Monitoring

- Continuous surveillance of sites in real-time
- Monitor multiple locations from a central location
- Quickly access cameras during alarms or events
- Operator workstations for active surveillance

Active Surveillance

- Actively control PTZ cameras to follow activity
- Monitor large areas with PTZ cameras
- Zoom in on specific areas or subjects
- Engage with subjects via audio if available

Situational Awareness

- Maintain awareness of overall site status
- Monitor multiple areas simultaneously
- Organize cameras by area, priority, or function
- Quick visual verification of site conditions

Mobile Monitoring

- Access live feeds from mobile devices
- Native mobile applications for iOS and Android
- Intuitive touch-based controls on mobile

Best Practices

Layout Organization

- Group cameras by location, function, or priority

- Place important cameras in prominent positions
- Organize for quick access to needed cameras

Performance Optimization

- Do not exceed system capabilities for concurrent streams
- Balance quality vs performance
- Enable Local Mode for on-site workstations when available
- Close streams when not needed

Security

- Use secure connections (HTTPS)
- Ensure proper user authentication
- Limit access based on user roles
- Monitor access to live view via audit logging

Limitations and Considerations

Network Requirements

- Requires adequate bandwidth for streaming
- Network latency affects real-time viewing
- Stable network connection required
- More streams require more bandwidth

Device Capabilities

- Devices have limits on concurrent streams
- Features depend on device protocol support and firmware
- Device hardware capabilities affect streaming quality